Turbulence & the Kolmogorov Spectrum: 0/0 at the Dissipation Scale

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The Removable Singularity Framework

Abstract

We show that turbulence has a 0/0 removable singularity at the Kolmogorov dissipation scale eta, where the inertial energy cascade meets viscous dissipation. The Kolmogorov 1941 (K41) energy spectrum E(k) = C_K * epsilon^{2/3} * k^{-5/3} has a universal exponent -5/3 that is a COMPLETELY DIFFERENT universality class from the Ising model. The Richardson cascade is a self-similar 0/0 at each scale. Intermittency corrections (She-Leveque 1994, mu = 2/9) modify the spectrum via a log-Poisson cascade. The Reynolds number transition Re_c ~ 2000-4000 is another 0/0. The -5/3 exponent appears in atmospheric turbulence, ocean currents, stellar interiors, galaxy formation, and QCD gluon cascades.

1. The Turbulence Problem

Turbulence is the last great unsolved problem of classical physics. Despite being observed for centuries (da Vinci drew turbulent eddies in 1508), a complete mathematical theory remains elusive. The Navier-Stokes millennium problem asks whether smooth solutions exist in 3D -- the answer is unknown.

Yet Kolmogorov (1941) discovered a UNIVERSAL law: the energy spectrum of turbulence follows E(k) ~ k^{-5/3} across ALL turbulent flows, from coffee cups to Jupiter. This is the most important universal law in classical physics after thermodynamics.

2. Kolmogorov 1941 (K41) Theory

Kolmogorov's two hypotheses of local isotropy and similarity:

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E(k) = C_K * epsilon^{2/3} * k^{-5/3}
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where C_K = 1.5 (Kolmogorov constant) and epsilon is the energy dissipation rate.

The -5/3 exponent is UNIVERSAL: it depends on nothing -- not viscosity, not geometry, not forcing.

3. The Dissipation Scale (0/0 Point)

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eta = (nu^3 / epsilon)^{1/4}
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Below eta: viscous dissipation converts kinetic energy to heat. Above eta: inertial cascade transfers energy to smaller scales. At eta: the energy transfer is 0/0 -- the cascade meets viscosity. This is the removable singularity of turbulence.

4. Richardson Cascade (1922)

"Big whorls have little whorls / Which feed on their velocity, / And little whorls have lesser whorls / And so on to viscosity."

- Energy cascades from large to small scales
- Self-similar 0/0 at each scale (fractal structure)
- Energy flux Pi(k) = epsilon = constant in inertial range
- At dissipation scale: Pi(k) -> 0 (removable singularity)

5. Reynolds Number Transition

$$Re = U*L/nu$$

Below Re_c ~ 2000: laminar flow (order parameter = 0). Above Re_c ~ 4000: turbulent flow (order parameter > 0). At Re_c: 0/0 removable singularity (transition to turbulence).

6. Intermittency & She-Leveque (1994)

K41 assumes Gaussian velocity statistics. Real turbulence has intermittency: rare intense events (e.g., extreme wind gusts).

$$zeta\_p = p/3 - mu*p*(p-3)/18 \quad \text{with } mu = 2/9$$

The She-Leveque model replaces the Gaussian cascade with a log-Poisson cascade. The intermittency parameter mu = 2/9 is UNIVERSAL across all turbulent flows.

7. Two Great Universalities

The 0/0 framework has TWO fundamental universalities:

System	Exponent	Class
K41 turbulence	-5/3	Kolmogorov
Ising 2D	1/8	Onsager
Ising 3D	0.326	Monte Carlo
Ising MF	1/2	Bragg-Williams
ER percolation 2D	1/3	Erdos-Renyi

K41 -5/3 is NOT an Ising class. It is a fundamentally different universality class, arising from energy conservation rather than symmetry breaking.

8. Conclusion

Turbulence has a 0/0 removable singularity at the dissipation scale. The -5/3 exponent is the most universal law in classical physics, appearing in atmospheric turbulence, ocean currents, stellar interiors, galaxy formation, and QCD. The 0/0 framework now has TWO fundamental universalities: Ising (phase transitions) and Kolmogorov (turbulence).

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